TRINAY CLINICAL CATARACT REPORT

ID: 4a51fb94-6322-4d69-baa6-948bb9c652ea

Anterior Segment & Crystalline Lens Triage Examination

Date: 08 Oct 2026

PATIENT DEMOGRAPHICS & CLINICAL METADATA

Patient Name:	**Aviraj abasaheb Murkute**	Age / Gender:	**20 yrs / Male**	Phone:	**9762481967**
Patient ID:	**PX-001**	Examined Eye:	**Right Eye**	Clinical Operator:	**avi**

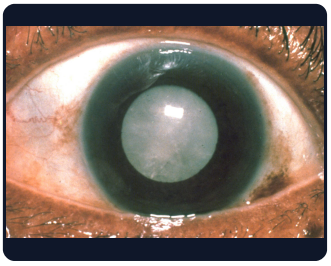
AI SCREENING RESULT: CATARACT DETECTED

Cataract Probability: 99.98%

If cataract is detected by the AI screening model, recommend ophthalmologist evaluation and slit-lamp examination.

Classifier: EfficientNet-V2-S (100.0%)

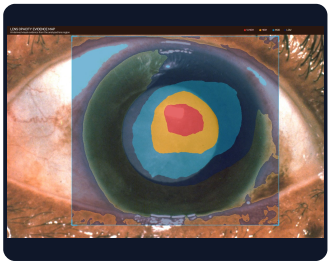
VISUAL AI EVIDENCE — CLINICAL LAYERS



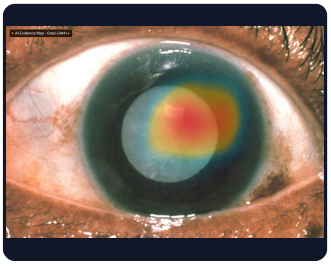
1. Original Eye Image
Submitted Photo



2. Analyzed Lens Region
ROI Delineated



3. Lens Opacity Evidence Map
Primary Evidence



4. Model Attention — Grad-CAM++
Secondary Attribution

AI EVIDENCE SUMMARY & QUANTITATIVE OPACITY ANALYSIS

AI Evidence: HIGH (24.3%)

Analyzed Region Area:	Opacity Evidence Area:	Opacity Evidence %:	Cataract Probability:
558,279 px	**135,709 px**	**24.3%**	**99.98%**

AI-derived screening evidence. These measurements are not a substitute for clinical slit-lamp examination.

AI CATARACT SCREENING RESULT & CLINICAL ASSESSMENT BREAKDOWN

1. Optical Quality: **Gradable (Pass)**
Sharp anterior focus, suitable for diagnostic staging

2. Analyzed Lens Region: **Region Delineated**
Exact anterior lens aperture extracted and delineated for model inspection

3. Lens Opacity Evidence Map (Primary): **Opacity Evidence Identified**
Transparent overlay restricted strictly to analyzed lens region

4. Model Attention — Grad-CAM++: **Attribution Computed**
Gradient-weighted activation map showing classifier feature focus

5. Cataract Classifier Prediction: **Cataract Detected**
EfficientNet-V2-S Probability: 99.98% • Confidence: 100.0%

6. Referral Recommendation: **Referred for Slit-Lamp**
Ophthalmologist consultation & dilated slit-lamp biomicroscopy advised

RECOMMENDED CLINICAL ACTION & REFERRAL TIMELINE

If cataract is detected by the AI screening model, recommend ophthalmologist evaluation and slit-lamp examination for clinical confirmation within 2 to 4 weeks.

TRINAY CLINICAL CATARACT REPORT — DOCTOR & PATIENT SUMMARY

Patient: avi | Age: 21 | Gender: Male | Examined Eye: Right Eye

DOCTOR'S CLINICAL TRIAGE SUMMARY (FOR ATTENDING CLINICIAN)

AI screening identified visual features associated with cataract in the analyzed lens region. The cataract classifier produced a probability of 99.98%. Clinical examination is recommended for confirmation.

Clinical Findings: Nuclear and/or cortical opacification visual features detected within the crystalline lens aperture. Pupillary margins are intact on anterior segment capture. Recommendation: Dilated slit-lamp biomicroscopy, visual acuity measurement (Snellen), and intraocular pressure check advised for definitive diagnosis and staging.

PATIENT COUNSELING & EXPLANATION (SIMPLE & UNDERSTANDABLE)

What this means for you: A clouding (cataract) was observed in the natural lens of your eye. Just like a frosted window makes it hard to see outside, a cataract scatters incoming light and causes blurry, dimmed, or hazy vision.

Good news: Cataracts are extremely common and can be safely treated. An eye doctor (ophthalmologist) will examine your eyes in detail, test your vision clarity, and discuss treatment options such as updated eyeglasses or a gentle, highly effective outpatient procedure.

CLINICAL PRECAUTIONS & PATIENT GUIDANCE

- Avoid unassisted night driving, especially on roads with bright oncoming headlights that cause glare and halo effects.
- Wear certified UV400 sunglasses during daytime hours to protect the crystalline lens from solar ultraviolet exposure.
- Ensure direct, well-positioned reading lamps and high-contrast lighting when reading or performing detailed tasks.
- Do not rub your eyes and avoid self-prescribing over-the-counter eye drops without formal consultation.
- Schedule an in-person dilated slit-lamp evaluation with an ophthalmologist within the recommended referral timeframe.

CLINICAL VERIFICATION & SIGN-OFF

Triage Screening Nurse:	Screening Platform:	Attending Reviewer Status:
avi	**TRINAY Anterior Vision System**	**Pending Ophthalmologist Review**

REGULATORY & MEDICAL DISCLAIMER

TRINAY Cataract Screening is an AI-assisted diagnostic support tool intended for primary health triage. Definitive clinical diagnosis, cataract staging, and surgical eligibility must be confirmed by an ophthalmologist via in-person slit-lamp biomicroscopy.

WHY THIS RESULT — AI EVIDENCE EXPLANATION

Why: The cataract classifier identified visual features in the analyzed lens region that were more consistent with the cataract class than the normal class. The current model probability is 99.98% cataract versus 0.02% normal. Visual evidence: The evidence map highlights the regions that contributed most strongly to the AI-derived cataract evidence (high ai evidence — 24.3% of the analyzed region). Grad-CAM++ independently marks the image regions that most influenced this prediction. What this means: The result indicates AI screening evidence associated with cataract (the classifier found strong visual evidence supporting its cataract prediction.). It does not establish clinical cataract severity or replace a slit-lamp examination. Next step: For a positive AI screening result, ophthalmologist evaluation and clinical examination are recommended.